

The 3 Ps of IBD today – prevention, pre-diagnosis and participation

Rupa Banerjee

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The year 2025 has seen a pivotal shift in inflammatory bowel disease research, from therapeutic management to prevention. Emerging evidence emphasizes the ‘3 Ps’ shaping future care – prevention, pre-diagnosis and participation – focusing on dietary risk modification, early biomarker detection and multisectoral engagement to advance proactive, population-level inflammatory bowel disease prevention.

Traditionally inflammatory bowel disease (IBD) was considered a disease of Western countries. Today, IBD has become a global disease, with rising incidence and prevalence across the highly populous low- and middle-income nations of the global south. Over the past few decades, IBD research was primarily focused on therapeutic advances to reduce inflammation, achieve mucosal healing and prevent complications in active disease. In parallel, there has been a growing interest in the reason for the rising cases of IBD, identifying pre-diagnostic indicators that precede disease onset and the need for early intervention. In 2025, key publications^{1–3} have emphasized the ‘3 Ps’ that will shape the future of IBD care: prevention, pre-diagnosis and multi-stakeholder participation (Fig. 1).

Prevention

The steadily rising incidence of IBD particularly in Asia has been attributed to environmental risk factors including westernization of lifestyles and alteration of dietary habits⁴. However, whether lifestyle and dietary modifications could prevent the development of IBD has been much less evaluated. Additionally, there are no standardized scoring systems to clearly define the beneficial or deleterious effects of a particular dietary pattern.

Several large-scale epidemiological studies have shown that a typical Western diet high in polyunsaturated fatty acids, red meat, refined sugars and ultra-processed food (UPF, with additives, preservatives and emulsifiers) and low in fibre, fruits and vegetables is associated with increased risk of developing IBD. These UPFs are thought to alter the function and composition of the gut microbiome and impair intestinal permeability, thereby promoting inflammatory pathways implicated in IBD⁵. In 2025, the Prospective Urban Rural Epidemiology (PURE) study by Narula et al.¹ has done a deep dive in this direction focusing on ultra-processed grains rather than all UPFs more broadly. Validated food frequency questionnaire-based data from a large cohort of 124,950 individuals from 21 countries collected over two decades clearly showed that higher consumption of ultra-processed grains was significantly associated with an increased risk of developing IBD.

Specifically, participants consuming ≥ 19 g per day of ultra-processed grains had a hazard ratio (HR) of 1.86 (95% CI 1.26–2.61) compared with those consuming < 9 g per day. Fresh bread and rice consumption (less processed or refined grains) were associated with lower IBD risk.

The study is unique for many reasons. First, it is a rare prospective epidemiological study including low- and middle-income countries covering both rural and urban population increasing generalizability beyond the so-called Western populations. Second, the prospective design allowed data collection before the onset of disease, reducing concerns of reverse causation. Importantly, quantifying ultra-processed grain intake provides specific insights into dietary risk factors that can be incorporated into dietary guidelines. Beyond individual dietary components, there is now growing evidence of a link between specific dietary patterns and development of IBD⁶. Notably, the Western dietary pattern increased while the Mediterranean diet reduced the risk of developing IBD⁵.

The year 2025 also saw a large prospective analysis from the European Prospective Investigation into Cancer and Nutrition (EPIC) cohort on dietary patterns and subsequent risk of incident Crohn’s disease and ulcerative colitis over long-term follow-up². In this study, an inflammatory score of the diet (ISD) was derived from validated country-specific food frequency questionnaires, based on the weighted associations of foods and nutrients with markers such as CRP, IL-6 and TNF. Among 394,255 participants across 10 European countries followed for a median of 13.6 years, higher ISD was significantly associated with an increased risk of Crohn’s disease (quartile 4 versus 1: adjusted HR 1.88; 95% CI 1.14–3.10; P -trend < 0.01), but not ulcerative colitis (adjusted HR 0.85; 95% CI 0.63–1.15). Low intakes of fibre, monounsaturated fatty acids, vitamin C, magnesium, onion and alcohol were identified as key contributors to higher inflammatory potential. These findings based on diverse dietary patterns together with a dietary score highlight the cumulative and interactive potential of foods habitually consumed. Interestingly, it is the everyday dietary patterns, rather than single foods

Key advances

- The PURE study showed that higher consumption of ultra-processed grains (≥ 19 g per day) was significantly associated with increased risk of developing inflammatory bowel disease; fresh bread and rice had lower risk¹.
- The EPIC cohort analysis demonstrated that a pro-inflammatory dietary pattern with a high inflammatory score correlated with an increased risk of Crohn’s disease².
- Distinct antibody responses to microbial antigens were identified at least 4 years before diagnosis of inflammatory bowel disease in a large population-based cohort; a Crohn’s disease-specific signature showed early reactivity to bacterial flagellins³.

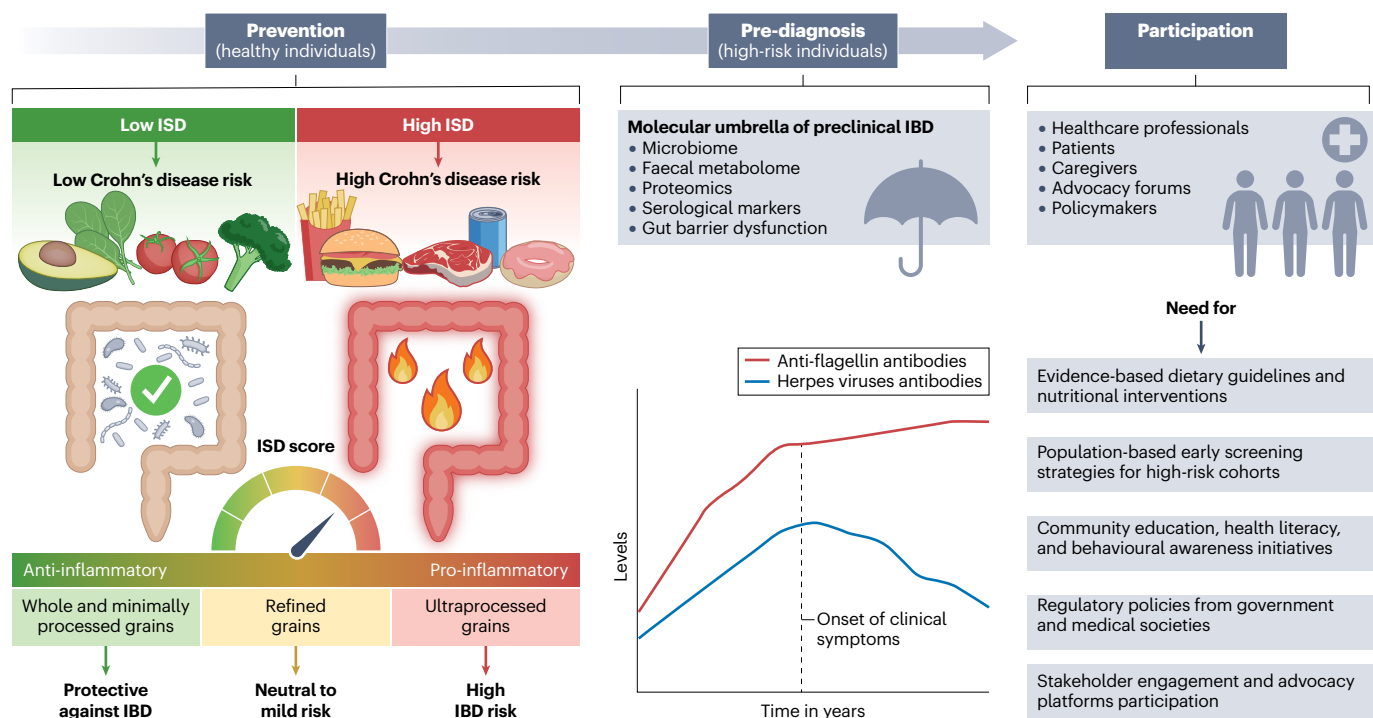


Fig. 1 | Trends in IBD care 2025: from prevention to early intervention. Overview illustrating emerging strategies in inflammatory bowel disease (IBD) care. Prevention: dietary patterns modulate intestinal inflammation and disease risk through the inflammatory score of diet (ISD). Pre-diagnosis:

molecular umbrella of preclinical IBD encompasses a range of factors, including microbiome and metabolome. Participation: integrating healthcare providers, patient advocacy forums and policymakers as multi stakeholders towards proactive, population-level disease prevention.

that are instrumental in influencing long-term systemic inflammation and future IBD risk. A scoring system could be a logical first step towards assessment of varied dietary patterns globally, forming the basis for preventive nutritional guidelines in a real-world scenario.

Pre-diagnosis

There is increasing evidence that IBD is preceded by an asymptomatic yet biologically active preclinical phase during which autoimmune and antimicrobial humoral immunity can be detected⁷. Large retrospective studies such as **PREDICT** and prospective analyses including the **Genetic, Environmental, Microbial (GEM) project** have consistently shown that serological changes can date more than a decade before the clinical onset of disease⁸.

A seminal study published in 2025 (ref. 3) has explored this preclinical window demonstrating not only the presence of early antibody signatures years before diagnosis, but also the changing antibody profile pre-diagnosis and post-diagnosis. A population-based cohort of 167,000 people was analysed, with 178 individuals with serum samples taken before and after IBD onset. Antibody responses to nearly 344,000 microbial, food and immune antigens were profiled using phage-display immunoprecipitation sequencing. Pre-diagnostic antibody signatures were dominated by bacterial flagellin antibodies primarily in Crohn's disease and continued to rise post-diagnosis. On the contrary, antibody responses against herpes viruses (Epstein–Barr virus, varicella zoster virus and herpes simplex virus 1) reduced post-diagnosis in both ulcerative colitis and Crohn's disease. The longitudinal sampling highlighted the immunological events leading to the development of IBD and enabled differentiation from multiple confounders, such as environmental exposures, transient infections or vaccination-induced immune responses. These results highlight the potential for predictive biomarkers to identify high-risk individuals and early interventions to prevent or modify IBD progression.

The preclinical phase of IBD, the typical 'lull before the storm', is now a priority area for further translational research. The umbrella for pre-diagnosis is multidimensional, including the genetic profile, immune dysregulation, gut microbial changes, altered intestinal permeability and the metabolome⁹. Artificial intelligence and machine learning-based algorithms will likely have a central role in integrating these complex datasets to develop accurate predictive models of disease onset.

Whether targeting these biomarkers would prevent the development of IBD will be the game-changer transitioning IBD care from reactive to proactive medicine.

Participation

The practical implementation of prevention and pre-diagnosis of IBD cannot be achieved without multi-stakeholder participation¹⁰. The complexity of disease determinants demands coordinated engagement across the scientific, clinical, public health and community domains. This work would involve public health awareness campaigns and integrating gut-health education into school curricula. Patient advocacy networks and social-media-driven initiatives can improve acceptance of screening tests in high-risk individuals and adherence to dietary guidelines. Governments and public-health agencies need to mandate clearer labelling of food processing and additive content, incentivize reformulation of industrial foods to reduce emulsifiers and preservatives along with imposing taxes and legal restrictions on UPFs.

Ultimately, participation will redefine IBD control as a shared societal goal rather than a specialist's burden alone.

Rupa Banerjee✉

IBD Centre, AIG Hospitals, Hyderabad, India.

✉ e-mail: dr_rupa_banerjee@hotmail.com

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Competing interests

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